

OpenAI

As a Product Analyst on the ChatGPT team, you are tasked with understanding how query complexity affects user engagement and system performance....

Question 1: What is the average response time for each query complexity level in January 2024?

```
SELECT complexity_level, AVG(response_time_seconds) AS average_response_time
FROM fct_queries
WHERE query_date BETWEEN '2024-01-01' AND '2024-01-31'
GROUP BY complexity_level
```



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Question 2: For each query complexity level, what is the average user satisfaction score for queries that took more than 2 seconds to respond in January 2024?

```
SELECT complexity_level, AVG(satisfaction_score) AS average_s  
atisfaction_score  
FROM fct_queries AS f  
INNER JOIN fct_user_engagement AS u  
USING (query_id)  
WHERE query_date BETWEEN '2024-01-01' AND '2024-01-31' AND res  
ponse_time_seconds > 2  
GROUP BY complexity_level
```



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Question 3: We want to identify the complexity level that optimizes both user engagement and system efficiency. Rank average user satisfaction (high to low) and average response time (low to high) across different complexity levels in January 2024. Which level has the best average satisfaction & response time ranking?

```
with total AS(
  SELECT complexity_level,
         AVG(response_time_seconds) AS average_response_time,
         AVG(satisfaction_score) AS average_satisfaction_score
  FROM fct_queries AS f
  INNER JOIN fct_user_engagement AS u
  USING (query_id)
  WHERE query_date BETWEEN '2024-01-01' AND '2024-01-31'
  GROUP BY complexity_level),

ranking AS(
  SELECT *,
         RANK() OVER(ORDER BY average_response_time) AS rank_response,
         RANK() OVER(ORDER BY average_satisfaction_score DESC) AS rank_score
  FROM total),

combined_rank AS (
  SELECT *,
         (rank_response + rank_score) AS combined_rank
  FROM ranking)

SELECT *
FROM combined_rank
ORDER BY combined_rank ASC
LIMIT 1
```

